Tectonophysics 510 (2011) 370–380

Contents lists available at SciVerse ScienceDirect

Tectonophysics

journal homepage: www.elsevier.com/locate/tecto

A paleomagnetic study from the Late Jurassic volcanics (155 Ma), North China: Implications for the width of Mongol–Okhotsk Ocean

Pei Junling [a,b,*], Sun Zhiming [a], Liu Jing [a,c], Liu Jian [a], Wang Xisheng [a], Yang Zhenyu [a], Zhao Yue [a], Li Haibing [b]

[a] Key Laboratory of Paleomagnetism, Institute of Geomechanics, CAGS, Beijing 100081, China
[b] State Laboratory for Continental Tectonics and Dynamics, Beijing 100081, China
[c] East China Mineral Exploration and Development Bureau for Non-ferrous Metals, Nanjing 210007, China

ARTICLE INFO

Article history:
Received 7 January 2010
Received in revised form 22 July 2011
Accepted 5 August 2011
Available online 17 August 2011

Keywords:
North China Block
Paleomagnetic
Mongol–Okhotsk Ocean
Siberia

ABSTRACT

A new paleomagnetic study has been carried out on the Tiaojishan Formation effusives in the Chengde area, North China Block (NCB). The age of the Tiaojishan Formation volcanics was recently dated from 153 Ma to 161 Ma by U–Pb and ^{40}Ar–^{39}Ar. A total of 154 samples from 16 sites within the Tiaojishan Formation were collected. Detailed stepwise thermal demagnetization allows us to isolate two components from 77 oriented drill-core samples. A low-temperature component identified in most samples falls close to the present local Earth field direction. Characteristic remanent directions obtained from high-temperature pass-reversal test yield a pole position at 240.3°E, 59.9°N with $\alpha 95 = 6.8°$. Combined with coeval paleopoles obtained from the Siberia Block (SIB), these results show a significant paleolatitude displacement between the NCB and SIB, which indicates that the Mongol–Okhotsk Ocean was still open with ~3000 km width in its eastern part at ca. 155 Ma.

© 2011 Elsevier B.V. All rights reserved.

1. Introduction

Existing paleomagnetic studies have demonstrated that the North China Block (NCB) and Amuria(AMU) should have formed a single entity since the Late Permian (Pruner, 1992; Zhang et al., 2007; Zhao et al., 1990), and then as one block completed the collision with the South China Block (SCB) during the Middle Jurassic to Late Jurassic (Enkin et al., 1992; Gilder and Courtillot, 1997; Lin et al., 1985; Ma et al., 1993; Yang and Besse, 2001; Yang et al., 1992; Zhao and Coe, 1987). However, the final amalgamation of this United East Asia Block (UEAB, including AMU, NCB and SCB) and the Siberia Block (SIB) is not well defined.

Jurassic–Cretaceous collision of an amalgamated NCB–AMU with SIB is widely believed to have accompanied closure of a Mongolo–Okhotsk Sea more than 800–1000 km north of the Yanshan Belt (Courtillot et al., 1994; Enkin et al., 1992; Nie, 1991; Nie and Rowley, 1994; Zhao et al., 1994; Zonenshain et al., 1990). The Mongol–Okhotsk suture zone (Fig. 1) lies within a geographically isolated region crossing southeastern Siberia and Amuria. It has been the focus of recent geologic and paleomagnetic investigations concerning the closure of an ocean (Cogné et al., 2005; Gilder and Courtillot, 1997; Halim et al., 1998; Kravchinsky et al., 2002; Zorin, 1999).

The most popular models suggested that the Mongol–Okhotsk Ocean has been closed successively from the west at the end of the Carboniferous to the east at the Early Cretaceous, based on the felsic intrusive ages along the Mongol–Okhotsk suture (Zonenshain and Kuzmin, 1997).

A recent paleomagnetic study from Late Jurassic to Early Cretaceous from both sides of the Mongol–Okhotsk suture (Cogné et al., 2005) and published paleomagnetic data (Halim et al., 1998; Kravchinsky et al., 2002) have showed that the Siberia Block and the UEAB have been accreted together by Early Cretaceous. According to the age of Taldan, Ingoda and Kremljevka effusives (118–128 Ma and 125–135 Ma, respectively), and the lack of reversed polarity within these localities, the Mongol–Okhotsk Ocean has closed at about 130 Ma (Cogné et al., 2005). Another 140–150 Ma pole obtained from the average results of Manzhouli, Hua'an (Zhao et al., 1990), Shadaron (in situ) and Tergen (in situ), suggests 1700–2700 km (±700 km) of northward movement of AMU (UEAB) with respect to Siberia since the Late Jurassic (Cogné et al., 2005). however, this 140–150 Ma age is dubious according to radiometric ages of effusives from Manzhouli and Hua'an, Inner Amuria, that range from 139 to 199 Ma (Zhao et al., 1990), as well as a possible early remagnetization of the Middle to Late Jurassic Shadaron and Tergen formations.

Clearly, due to the lack of paleomagnetic data with accurate age from the Late Jurassic to 120 Ma in this area, and to the still debated ages between Jurassic–Cretaceous boundary and M0 (the bottom of

* Corresponding author at: Key Laboratory of Paleomagnetism, Institute of Geomechanics, CAGS, 11#Minzu Xueyuan Nanlu, Beijing, China, 100081. Tel.: +86 10 68422365; fax: +86 10 68422326.
E-mail address: peijl@yahoo.com.cn (P. Junling).

0040-1951/$ – see front matter © 2011 Elsevier B.V. All rights reserved.
doi:10.1016/j.tecto.2011.08.008

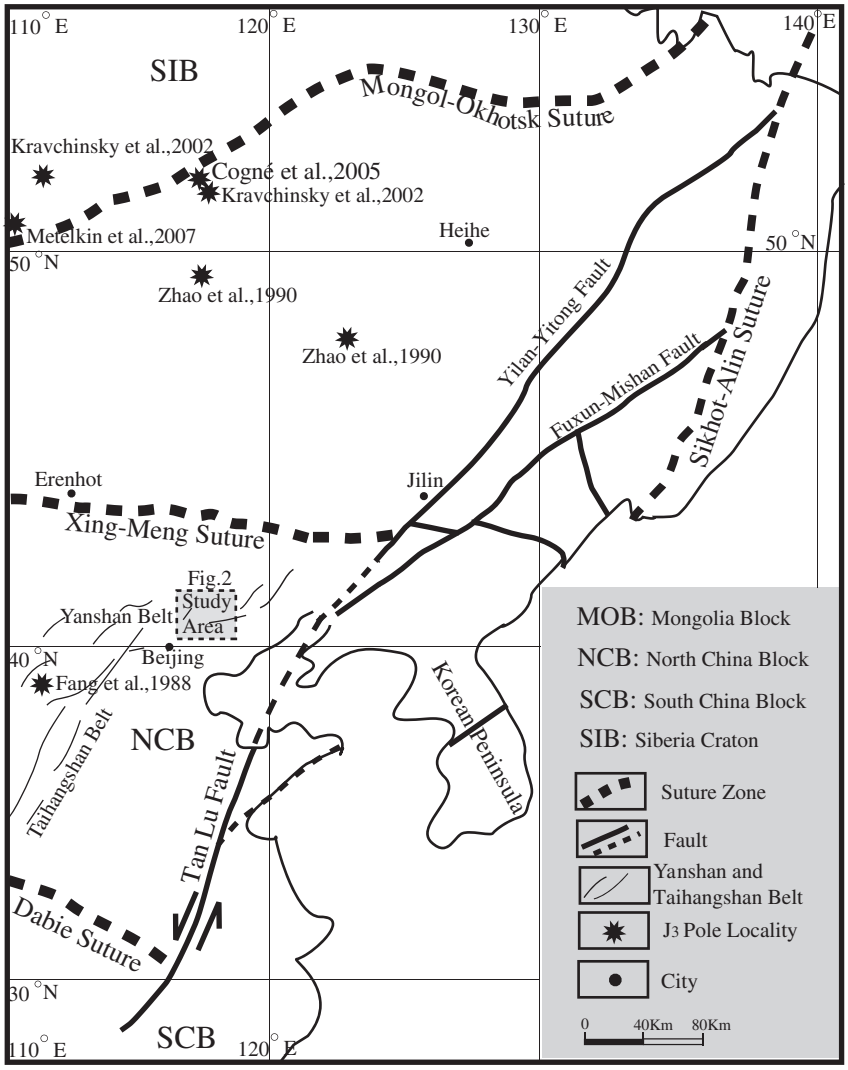


Fig. 1. Simplified tectonic map of East Asia. AMU, NCB, SCB, YSB, and SIB stand for Amuria, North China Block, South China Block, Yanshan Belt and Siberia, respectively.

Cretaceous Normal Superchron) (Channell et al., 1995; Gradstein et al., 2004; He et al., 2008; Ogg et al., 2004), the closure progress and precise age of the Mongol–Okhotsk Ocean remains uncertain.

Thus, new paleomagnetic results, particularly, Late Jurassic paleomagnetic poles with precise age, are crucial to understand the closure history of the Mongol–Okhotsk Ocean. A Late Jurassic (150–160 Ma) volcano sedimentary sequence (Ichetui Formation) filling up a series of grabens north of the Mongol–Okhotsk suture has been published (Metelkin et al., 2007). Here, we report a new ca.155 Ma (U–Pb and ^{40}Ar–^{39}Ar) paleomagnetic results from the Chengde area for the NCB, which provides new chronological and paleomagnetic constraints on the paleo-position of the NCB (UEAB) and the evolution of Mongol–Okhotsk Ocean.

2. Geological setting and paleomagnetic sampling

The Chengde basin is geologically located in the southern margin of the Yanshan Belt (YSB), itself located at the northern margin of the NCB (Fig. 1; Davis et al., 2001; Zhao, 1990). To the north, the Xing-Meng suture (Suolon suture) indicates a final closure of the Paleo-Asian ocean and the collision between the NCB and AMU during the Permian (Badarch et al., 2002; Li, 2006; Xiao et al., 2004; Yin and Nie, 1996).

The YSB experienced vertical movements accompanied by intense deformation during the Mesozoic, including multiple generations of folds, thrusts and extensional faults (Davis et al., 2001; Zhao, 1990). In the Early Jurassic, a series of rift basins bounded by normal faulting were developed on the amalgamated NCB and AMU (Darby et al., 2001; Meng, 2003; Zhao et al., 2002).

From Early Jurassic to Early Cretaceous, extensive terrestrial volcanic and clastic sediments were deposited in these fault-related rift basins. These volcanic rocks, varying from mafic to intermediate-acid in composition, were genetically related to the process of lithospheric replacement beneath the NCB (Zhang et al., 2003). The Tiaojishan flood basalts in the Chengde basin are the main focus of this paper.

This Late Jurassic volcanic rocks in the Chengde basin offer a unique opportunity to obtain key paleomagnetic results with accurate dating, and to constrain the tectonic evolution of the UEAB and adjacent areas. The most complete Tiaojishan Formation is well exposed along the road from Chengde to Xiabancheng with simple tectono-sedimentary setting (Fig. 2). The volcanic-rich Tiaojishan Formation has been assigned a Middle Jurassic age (Hebei Bureau of Geology, 1989), or an early Late Jurassic age (Beijing Bureau of Geology and Mineral Resources(BBGMR), 1991; Chen, 1996; Cui, 1999).

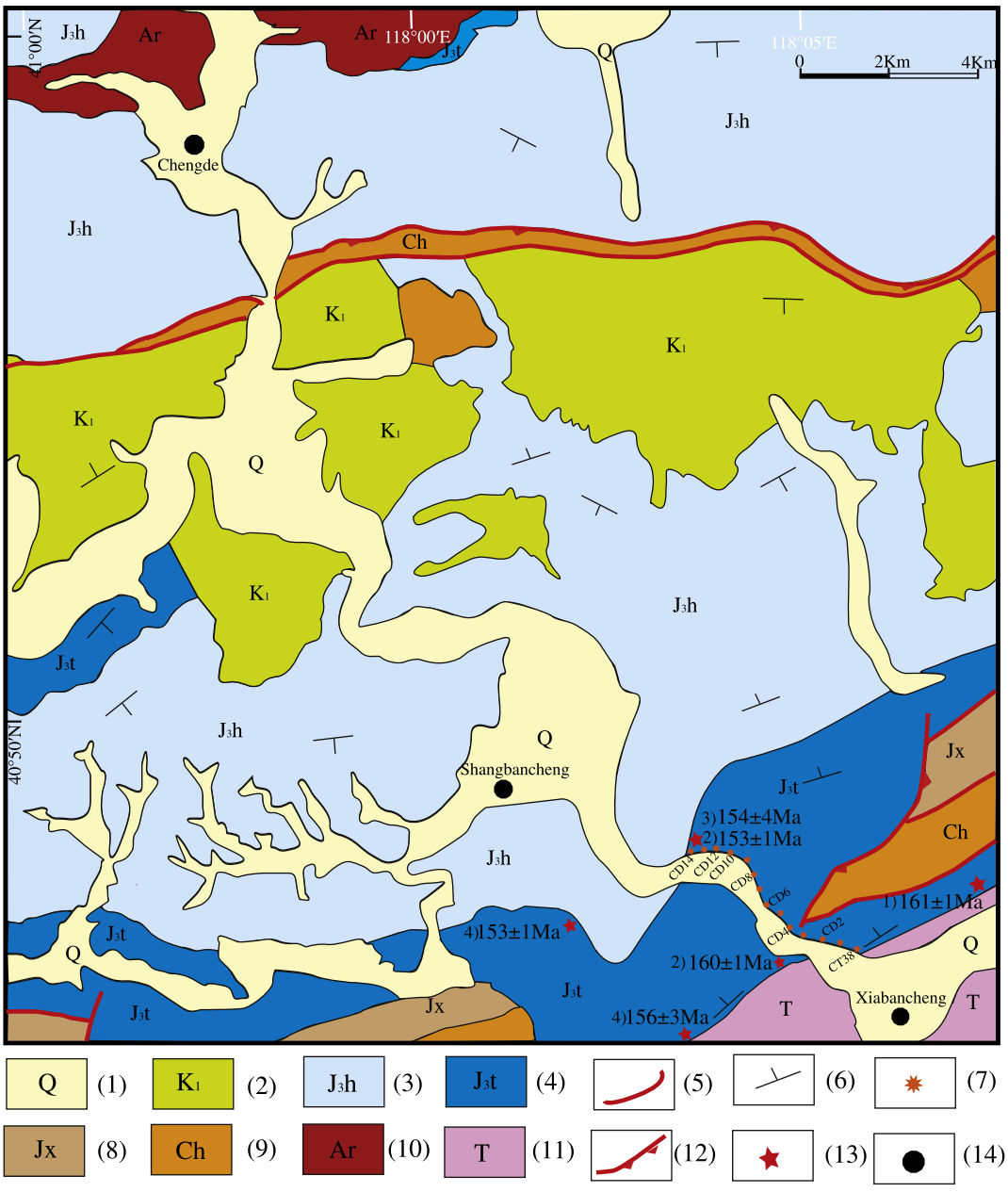


Fig. 2. Simplified geological and paleomagnetic sampling maps of the Chengde area in the NCB. (1) Quaternary; (2) Early Cretaceous; (3) Late Jurassic Houcheng Formation; (4) Late Jurassic Tiaojishan Formation; (5) Faults; (6) Strike and dip; (7) Location of our sampling sites; (8) Proterozoic Jixian Group; (9) Proterozoic Changcheng Group; (10) Archean; (11) Triassic; (12) Thrusts; (13) Location of published isotope ages of the Tiaojishan Formation, 1) Davis et al., 2001; 2) Cope, 2003; 3) Davis, 2005; 4) Liu et al., 2006; (14) City.

In recent years, major advancements in the understanding of the radiometric ages and geochemistry of the Tiaojishan Formation volcanic rocks in the Chengde basin have been rapidly achieved (Table 1; Fig. 2; Cope, 2003; Davis, 2005; Davis et al., 2001; Liu et al., 2006). Several U–Pb and ^{40}Ar–^{39}Ar ages have been obtained from the volcanic strata of the Tiaojishan Formation, ca.5 km north of Xiabancheng, and south of Chengde in Hebei province. The latest single-zircon U–Pb dating results show that the age of crystal tuff at

Table 1
Ages of the Tiaojishan Formation in Chengde basin.

Formation	Lithology (Mineral)	Age (Ma)	Method	Reference
Lower of Tiaojishan	Lithic tuff (biotite)	160.7 ± 0.8	^{40}Ar–^{39}Ar	Davis et al., 2001
Bottom of Tiaojishan	Andesite (hornblende)	159.7 ± 1.3	^{40}Ar–^{39}Ar	Cope, 2003
Top of Tiaojishan	Crystal tuff (sanidine)	152.6 ± 0.3	^{40}Ar–^{39}Ar	Cope, 2003
Top of Tiaojishan	Crystal tuff(zircon)	153.3 ± 3.3	U–Pb (SHRIMP)	Cope, 2003
Top of Tiaojishan	Andesite (zircon)	153.6 ± 3.8	U–Pb (SHRIMP)	Davis, 2005
Top of Tiaojishan	Lithic tuff (zircon)	152.5 ± 0.5	U–Pb (LA-ICP-MS)	Liu, et al., 2006
Bottom of Tiaojishan	Trachyandesite (zircon)	155.5 ± 2.9	U–Pb (LA-ICP-MS)	Liu, et al., 2006

Table 2
Sampling localities, geological suites, and description of local geology.

Site	Location	N (n)	Description of strata and local geology		
	$S_{lat.}$ (°N)/$S_{long.}$ (°E)		Formation	Lithology	Strike/dip
CT 38	40.793°N,118.110°E	10 (10)	Tiaojishan	Andesite	241/61
CD 1	40.793°N,118.110°E	10 (6)	Tiaojishan	Andesite	241/61
CD 2	40.793°N,118.110°E	11 (4)	Tiaojishan	Andesite	241/61
CD 3	40.793°N,118.110°E	10 (7)	Tiaojishan	Andesite	241/61
CD 4	40.793°N,118.110°E	10 (7)	Tiaojishan	Trachyandesite	241/61
CD 5	40.793°N,118.110°E	10(8)	Tiaojishan	Trachyandesite	241/61
CD 6	40.793°N,118.110°E	11 (5)	Tiaojishan	Trachyandesite	241/61
CD 7	40.794°N,118.109°E	8 (2)	Tiaojishan	Crystal tuff	241/61
CD 8	40.797°N,118.107°E	10 (5)	Tiaojishan	Crystal tuff	241/61
CD 9	40.801°N,118.102°E	9 (3)	Tiaojishan	Crystal tuff	241/61
CD 10	40.801°N,118.102°E	10 (4)	Tiaojishan	Crystal tuff	241/61
CD 11	40.801°N,118.102°E	8 (0)	Tiaojishan	Crystal tuff	241/61
CD 12	40.801°N,118.102°E	8 (3)	Tiaojishan	Crystal tuff	241/61
CD 13	40.801°N,118.102°E	10 (5)	Tiaojishan	Crystal tuff	241/61
CD 14	40.802°N,118.095°E	10 (7)	Tiaojishan	Crystal tuff	241/61
CD 15	40.802°N,118.092°E	9 (1)	Tiaojishan	Lithic tuff	241/61

Strike/dip = strike azimuth and dip of bed; N (n) = number of total samples collected (number of samples demagnetized); Slat./Slong. = north latitude/east longitude of sampling sites.

the top of the Tiaojishan Formation is about 153 ± 1 Ma, and the age of trachyandesite at the bottom is 156 ± 3 Ma in the Chengde basin (Table 1; Fig. 2; Liu et al., 2006).

We sampled Jurassic volcanic rocks along a road from Xiabancheng to Chengde (Fig. 2). The Tiaojishan Formation mainly consists of predominantly massive volcanic–clastic sedimentary breccias, mafic lavas, lithic tuffs, and water-lain tuffaceous sediments (BBGMR, 1991). The Tiaojishan Formation is conformably or disconformably underlain by the Jiulongshan Formation which contains many fossils (mainly *Ciniopteris phoenicopsis*) and is conformably overlain by the Houcheng Formation (t abundant fossils, *Djungarica–Mantelliana–Damonella* (BBGMR, 1991)).

A total of 154 samples from 16 sites within the Tiaojishan Formation were collected using a water-cooled portable drill and orientated using a magnetic compass. Accordingly, field measurements were not corrected for local magnetic declination (−7.3°) computed from the IGRF (2009) at our site locations. Eight to eleven cores were taken at each site representing a stratigraphic thickness of about 10 m. The tilt of the Tiaojishan Formation was measured only at two sites in the field because of poor outcrop conditions.

In the Paleomagnetic Laboratory of the Institute of Geomechanics in Beijing, one or two, 2.2 cm-long with 2.5 cm diameter cylindrical specimens were sliced from each sample with a diamond saw for rock magnetic and paleomagnetic measurements. Unfortunately, some samples were damaged and some specimens were fractured during sample processing, and only 77 specimens remained (Table 2). Because these volcanic specimens have high NRM intensities, a JR-6 spinner magnetometer must be used to measure remanent magnetization. However, samples with cracks cannot be tested using a JR-6 spinner magnetometer for its high-speed rotation.

3. Paleomagnetic results

3.1. Treatments

In order to help identify the magnetic minerals within the samples, both acquisition of Isothermal Remanent Magnetization (IRM) and three-axis thermal demagnetization of IRMs (Lowrie, 1990) were conducted in the Paleomagnetic Laboratory of the Institute of Geomechanics in Beijing. Fields of 2.4 T, 0.4 T and 0.12 T were applied along the Z, Y and X axes of the specimens, respectively. Then, stepwise thermal demagnetization up to 680°C was performed. IRM were measured using the IM-10 impulse magnetizer and the JR-6 magnetometer.

All samples were subsequently subjected to progressive thermal demagnetization up to 600 °C or 680°C in 12–14 steps using an ASC-Scientific TD-48 thermal demagnetizer with an interval residual field lower than 10 nT. The Natural Remanent Magnetization (NRM) were measured on a JR-6 spinner magnetometer due to their high NRM intensities. Paleomagnetic directions of the specimens were analyzed by principal component analysis (Kirschvink, 1980). Site-mean and overall-mean directions were computed using Fisher statistics (Fisher, 1953). All the interpretations and data processing have been done using the *PaleoMac* software package developed by Cogné (2003).

3.2. Rock magnetic results

Rock magnetic experiments are designed to identify the magnetic minerals within the rocks. Acquisition and opposite field demagnetization behavior of IRM are observed for selected specimens.

In some specimens, IRM curves show that saturation is not reached after applying a field of 2.4 T and a quasi-saturation (>80%) behavior at about 1.1 T (Fig. 3a). Combining remanent coercivity (Hcr>500 mT) deduced from back-demagnetization with three-component IRM thermal demagnetization, the presence of high-coercivity hematite minerals in these samples is highlighted. Indeed, the hard component (Z) unblocks at about 680 °C, typical for hematite. The soft component is unblocked around 580 °C (Fig. 3b), suggesting the presence of magnetite. In most specimens, however, saturation is reached more rapidly, after applying a lower field of about 0.4 T (Fig. 3c, e). Combining remanent coercivity (Hcr<50 mT) deduced from back-demagnetization with three-component IRM thermal demagnetization, the presence of low-coercivity magnetite minerals in these samples is demonstrated. All the components are unblocked around 580 °C (Fig. 3d, f), suggesting the presence of magnetite. These experiments indicate that magnetite is the main magnetic carrier in the Tiaojishan Formation, but hematite probably exists in some parts of the rocks.

3.3. Magnetic component analysis

Seventy-seven specimens were measured using thermal demagnetization with temperature intervals ranging from maximum 80 °C at the lower temperature to minimum 10 °C at the high temperature. Two NRM components are recognized in the samples (Table 3). We did not correct the local magnetic field declination in the field, therefore a magnetic declination (−7.3°) computed from IGRF (2009) at sample locations, was corrected for in the data. Because of the lack of data (only 1 or 2) for calculating a component in each site (e.g., CD1, CD2, CD6, CD7, etc.), we have merged two sites to improve the statistics.

Component A can be identified between NRM and 280 °C in most specimens, with a northerly declination and a steep positive inclination *in-situ* (Fig. 4). The precision parameter (k) is similar *in-situ* and *tilt-corrected* populations, mainly because of the monoclinal strata. However, the site-mean direction (Dec = 12.5°, Inc = 57.9°, $\alpha 95 = 3.2°$) prior to bedding correction is indistinguishable from the present dipole field that is Inc = 59.2° (Fig. 5a, b and Table 3), and is therefore probably a viscous remanent magnetization.

Most specimens demonstrate only a Component A (Fig. 4a), a higher temperature Component B of both reversed and normal polarities is isolated up to 600 °C or 680 °C in two thirds of the samples (Fig. 4b–i), which decays linearly to the origin for two thirds of the samples. Most sites have a reverse polarity with the exception of sites CD12–13 and CD14–15, which exhibit normal polarity.

The average site-mean direction is D = 267.7°, I = −43.5°, $\alpha_{95} = 7.3°$ prior to, and D = 207.1°, I = −37.9° and $\alpha_{95} = 7.5°$ after tilt corrections (Table 3, Fig. 5c, d). The fold test is inconclusive at the 99% and 95% probability level according to both tests of McElhinny

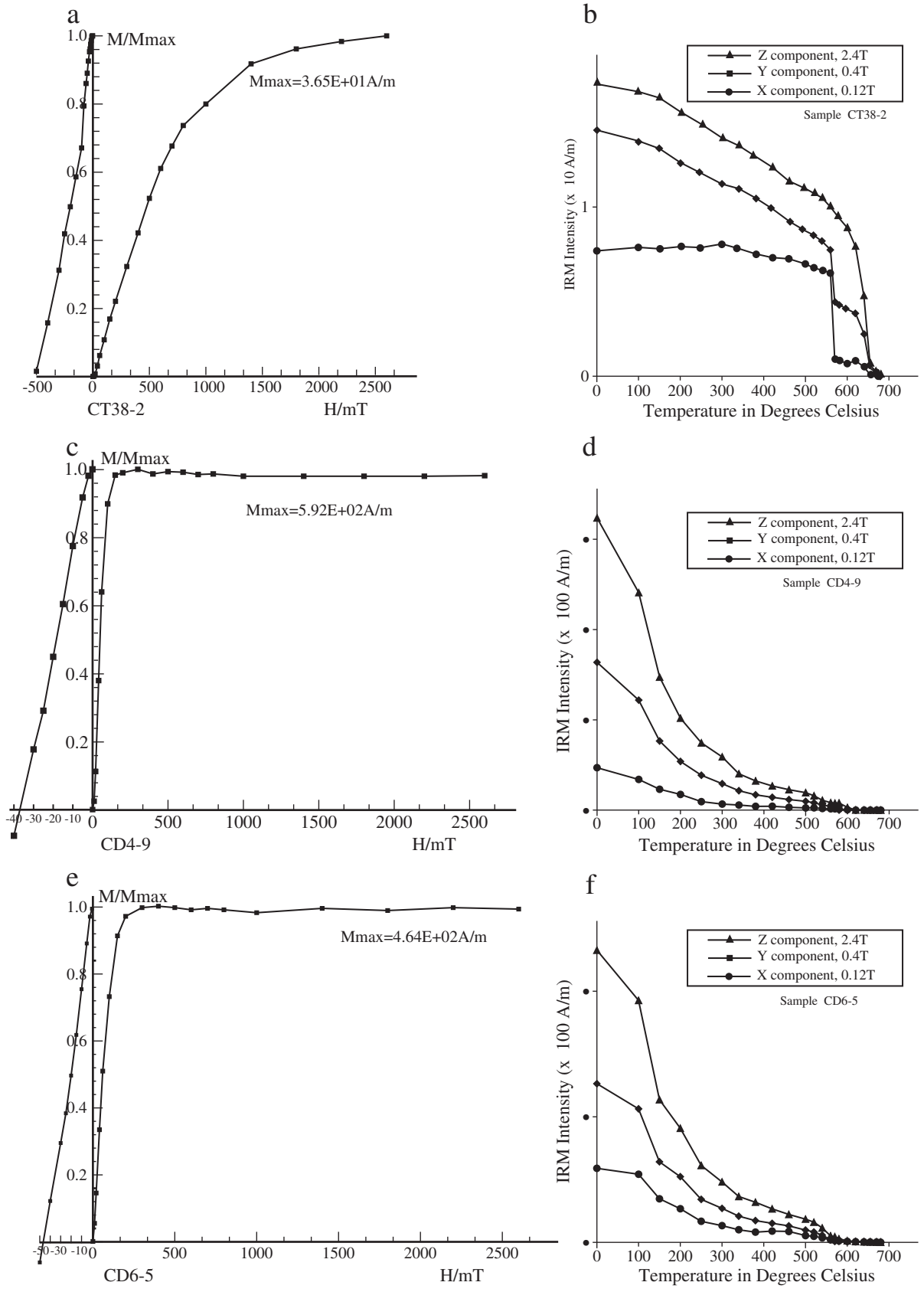


Fig. 3. Normalized Isothermal Remanent Magnetization (IRM) acquisition curves and opposite field demagnetization of representative samples (a, c, e); three-axis IRM thermal demagnetization showing unblocking temperatures around 680 °C or 580 °C (b, d, f).

Table 3
Site-mean directions and corresponding poles from Tiaojishan Formation in Chengde area, North China.

Site	n/N_0	N/R	Dg	Ig	Ds	Is	*k*	α_{95}	$P_{lat.}$	$P_{long.}$
Lower-temperature Component A										
CT38	10/10	10/0	10.6	60.2	349.5	5.0	170.2	3.7		
CD 1–2	10/10	10/0	6.0	58.4	348.4	2.4	229.0	3.2		
CD 3	6/7	6/0	33.5	62.5	356.0	14.1	11.7	20.4		
CD 4	7/7	7/0	12.4	59.8	350.4	5.3	47.0	8.9		
CD 5	8/8	8/0	9.6	54.7	352.2	0.0	114.4	5.2		
CD 6–7	6/7	6/0	17.6	58.7	354.8	5.0	28.6	14.6		
CD 8	4/5	4/0	9.6	55.6	352.5	0.3	64.4	11.5		
CD 9–10	7/7	7/0	5.7	59.2	348.0	2.8	63.5	7.6		
CD 12–13	7/8	7/0	13.1	58.2	352.4	3.6	33.5	10.6		
CD 14–15	8/8	8/0	10.7	50.1	355.2	−3.4	17.7	13.5		
Mean	10	–	12.5	57.9	–	–	232.0	3.2		
		–	–	–	351.9	3.5	224.7	3.2		
Higher-temperature Component B										
CT38	10/10	0/10	250.7	−39.7	207.8	−25.1	6.9	19.8		
CD 1–2	0/10	0/0								
CD 3	7/7	0/7	268.1	−52.2	195.4	−39.1	73.1	7.1		
CD 4	7/7	0/7	292.2	−50.9	192.9	−53.9	36.8	10.1		
CD 5	8/8	0/8	272.0	−40.5	210.9	−41.1	24.9	11.3		
CD 6–7	7/7	0/7	269.4	−45.6	205.4	−38.6	75.4	7.0		
CD 8	5/5	0/5	267.3	−26.1	229.0	−32.5	76.2	8.8		
CD 9–10	7/7	0/7	253.5	−42.2	205.8	−27.7	26.9	11.9		
CD 12–13	7/8	7/0	94.7	44.8	26.6	42.4	79.0	6.8		
CD 14–15	8/8	7/1	86.3	44.5	25.2	37.3	51.2	7.8		
Mean	9	–	267.7	−43.5	–	–	50.5	7.3		
		–	–	–	207.1	−37.9	48.2	7.5	59.9	240.3

n/N0 = number of samples used to calculate mean/total samples demagnetized; N/R = number of samples exhibiting positive/negative polarity; Dg/Ig (Ds/Is) = declination and inclination in situ (after tilt correction); Plat./Plong. = north latitude/east longitude of virtual geomagnetic pole; k = best estimate of the precision parameter; α95 = 95% confidence of the mean direction pole. McFadden and McElhinny (1990) reversal test, angle between the mean directions $\gamma = 2.8° < \gamma$ critic = 19.4° indicates a C class result.

(1964) and Watson and Enkin (1993). This is not surprising, considering the monotonous angle between the tilt of the different sites. Component B directions pass the reversal test of McFadden and McElhinny (1990) at the 95% probability level (C class) with an angle between the mean directions $\gamma = 2.8° < \gamma_{critic} = 19.4°$.

4. Discussion and preliminary conclusions

It is important to determine a paleomagnetic pole from reliable dating when used to study plate tectonics. We have used published isotopic dates from U–Pb and ^{40}Ar–^{39}Ar dating methods to get a well-defined age of the Tiaojishan Formation volcanics in the Xiabancheng section (Table 1; Fig. 2). The age of the Tiaojishan Formation is constrained by U–Pb and ^{40}Ar–^{39}Ar ages from 158 ± 1 Ma to 153 ± 1 Ma (Liu et al., 2006), allowing to reasonably assign an age of 155 Ma to the paleomagnetic pole obtained from the Tiaojishan Formation volcanics.

According to magnetic mineralogical studies, magnetite with low-coercivity and 580 °C unblocking temperature is the major remanent carrier in most of the measured samples except for site CT38, where hematite with both high-coercivity and high-unblocking (680 °C) temperature is the major remanent carrier in these samples. Moreover, most of the maximum unblocking temperature is 600°C (Fig. 4b, c, e, g, h, i). The HTC may be carried by magnetite and a small amount of hematite in these higher unblocking temperature samples. Three-component IRM thermal demagnetization curves (Fig. 3d, f) show that 90% drop at about 580 °C, the Curie point of magnetite, and 100% at 600 °C.

The sampling area is located in a simple fold belt and no major metamorphic events are known to have affected the region. The high-temperature characteristic remanent magnetization (ChRM) passes reversal tests, implying that the Component B is acquired during a long-time instead of in a short-time remagnetization event. In addition, corresponding paleomagnetic pole at 240.3°E, 59.9°N ($\alpha_{95} = 6.8°$) is completely different from the post-Late-Jurassic paleopoles (Fig. 5; Gilder and Courtillot, 1997; Yang et al., 1992). Therefore, the Component B, with positive reversal test, and a significant difference from the post-Late-Jurassic poles of the North China Block, probably represents the primary remanent directions.

Besides this paleopole, there are a few other Middle and Late Jurassic paleopoles obtained from surrounding blocks, as well as the Siberian poles of Metelkin et al. (2007) and Kravchinsky et al. (2002). In the following, the reliability of these data was analyzed, then, the relationships of our results and above coeval poles (listed in Table 4 and illustrated in Fig. 6) were discussed. Finally, the width of the Mongol–Okhotsk Ocean at 155 Ma was estimated, based on our new results and previously published data.

4.1. Analysis of paleopoles from surrounding blocks

Paleomagnetic data from the Middle and Late Jurassic have been published on the NCBand SCB (Table 4). However, few paleomagnetic data have been studied using modern instruments and data analysis techniques. In addition, there are no paleomagnetic data with reliable dating. In our compilation of the Middle Jurassic poles from the NCB, we follow Yang et al. (1992) and Gilder and Courtillot (1997) by omitting the Middle Jurassic results from Cheng and Fang (1980), Lin (1984), Fang et al. (1988), and Zheng et al. (1991), because these studies are based on just a few sites, or no details on demagnetization methods and stability tests were available.

Some Late Jurassic paleomagnetic data from the SCB have been obtained (Table 4) and are considered reliable (Gilder and Courtillot, 1997; Yokoyama et al., 2001). We have recomputed an average Late Jurassic pole from these data, which is consistent with the NCB Late Jurassic pole (Fig. 6; Gilder and Courtillot, 1997), indicating that the collision between the NCB and SCB was complete by the Late Jurassic.

Recently, paleomagnetic data of basalts from Transbaikalia (north of the Mongol–Okhotsk suture) are considered as a key paleopole for

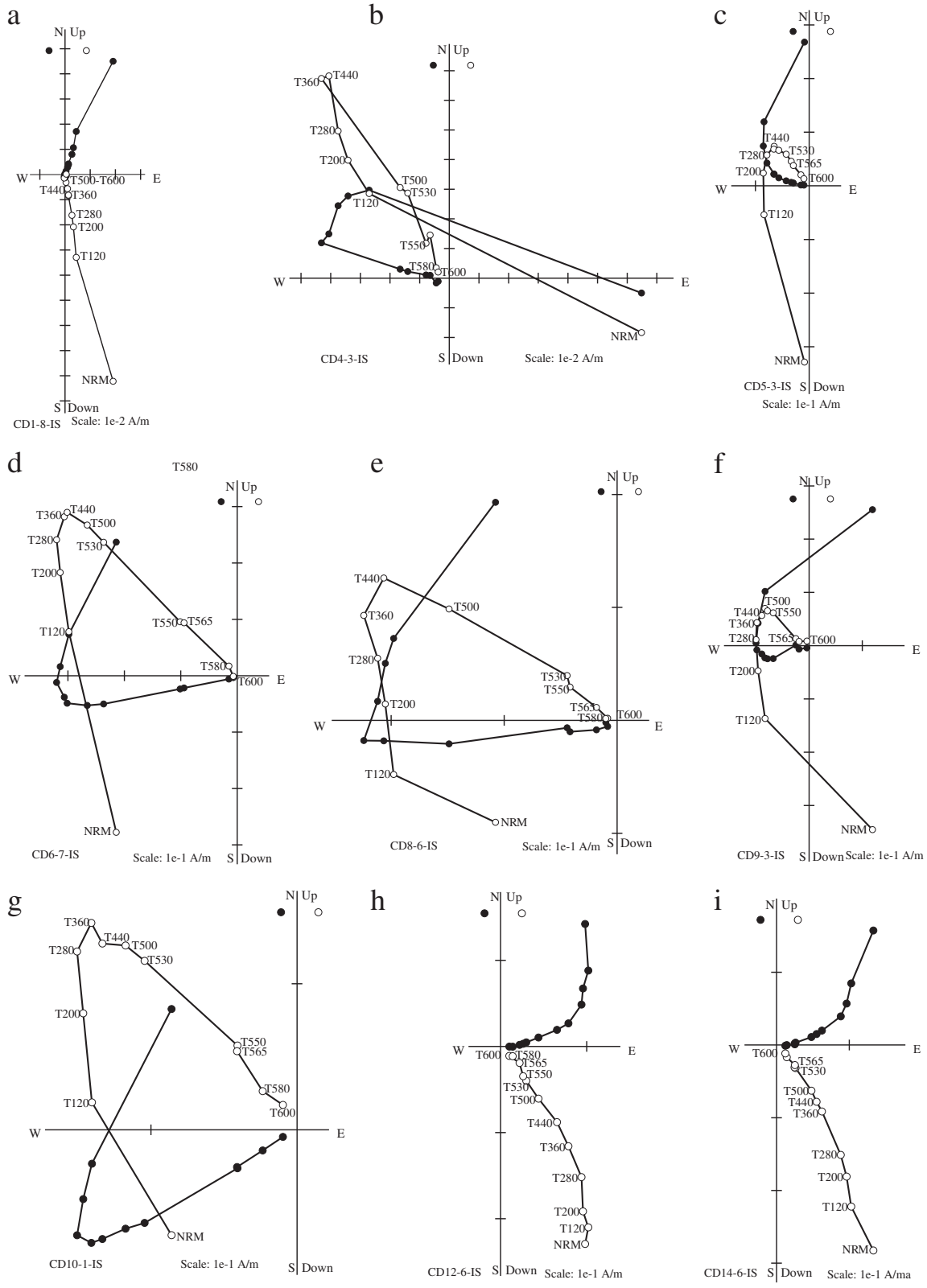


Fig. 4. (a to i) Selected Zijderveld diagrams of thermal demagnetizations of NRM in situ showing one component (a) and two components (b–i). Solid (open) symbols refer to the projection on the horizontal (vertical) plane. T200 = 200 °C.

the Siberia Block at 155 Ma (Metelkin et al., 2007). The age has been determined using the Rb–Sr and K–Ar isotopic methods. The mean direction is D = 44.4°, I = 71.7° and $\alpha 95 = 5.2°$ after tilt correction, yielding a paleopole at 166.8°E, 63.6°N with $\alpha 95 = 8.5°$. This pole coincides with the Jurassic result (161.0°E, 64.4°N with $\alpha 95 = 7.0°$) obtained by Kravchinsky et al. (2002) from the 150–160 Ma-old volcano sedimentary sections of the Badin Formation of the Chita region (Fig. 1; north of the Mongol–Okhotsk suture). Both the

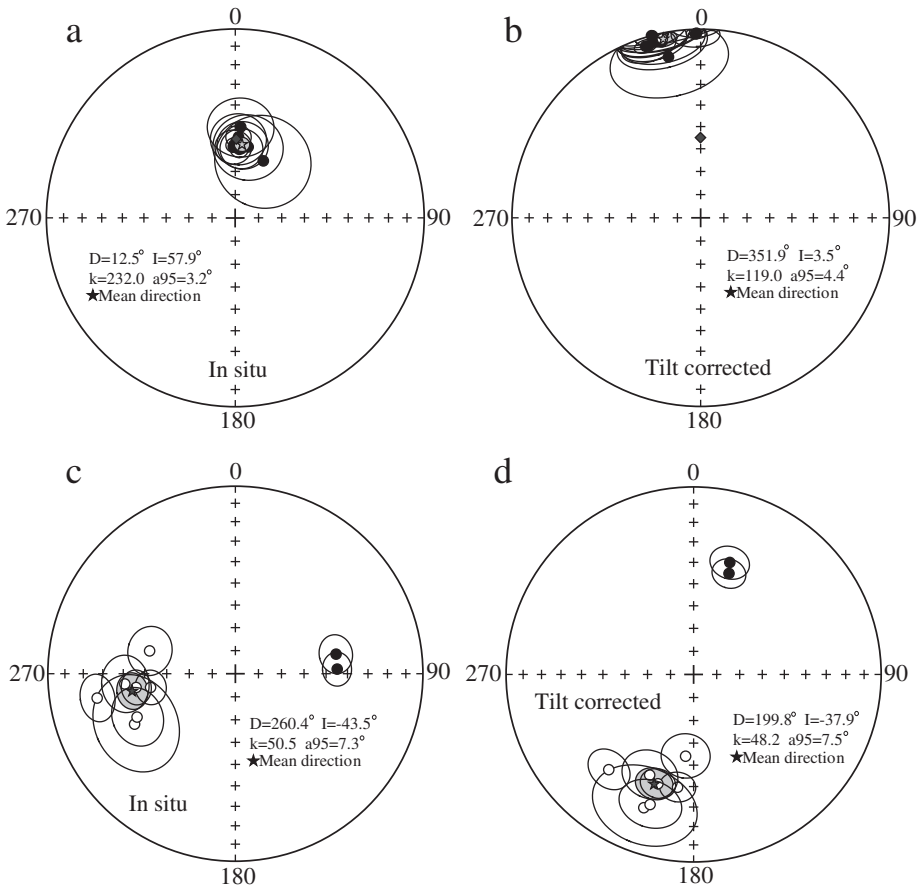


Fig. 5. Equal-area stereographic projections of Component A site-mean (a) prior to and (b) after tilt correction. (c, d) Site-mean directions of Component B. Lower (upper) hemisphere directions are marked with solid (open) symbols. Open circles represent the 95% confidence of the mean direction pole.

paleopoles of Metelkin et al. (2007) and Kravchinsky et al. (2002) are consistent with reference APWP at 150–160 Ma with a slight clockwise rotation (Fig. 6).

Coeval paleomagnetic poles (Table 4; Cogné et al., 2005; Kravchinsky et al., 2002; Zhao et al., 1990) from the AMU (Fig. 1; south of the Mongol–Okhotsk suture) are consistent with the NCB's Middle Jurassic pole (Fig. 6).

4.2. Discussion of ca.155 Ma paleopoles

In Fig. 6, we show all poles found in previous studies, together with our new results and related APWP of Eurasia. It is clear that our 155 Ma pole is consistent with AMU, NCB and SCB, all being far-sided from the 140–160 Ma part of the reference APWP (Fig. 6). However, our 155 Ma pole is closer to that of AMU than NCB and SCB.

Interpreting this difference is not straightforward. However, poles of the reference APWP undergo a >15° rapid "northward shift" between 140 and 160 Ma when viewed from site location (NCB + SCB). This shift is identical to the order of magnitude of the discrepancy between our 155 Ma pole and the NCB + SCB poles. Therefore, because of age uncertainties on sedimentary formation paleopoles, this difference can be interpreted as this uncertainty in average paleopoles of the NCB + SCB and this rapid shift of reference pole.

Our pole position differs slightly from that of published poles coming from the Middle and Late Jurassic of the NCB and SCB (Fig. 6) primarily because there are no enough reliable data in the comparison between different age poles. Thus, our well-dated 155 Ma pole supplied a key pole to investigate the history of the Mongol–Okhotsk Ocean closure.

4.3. Width of the Mongol–Okhotsk Ocean at 155 Ma

Fortunately, key paleopoles of the Siberia Block at 155 Ma have been published (Kravchinsky et al., 2002; Metelkin et al., 2007). In terms of rotation of the NCB with respect to SIB, our 155 Ma pole exhibits negligible rotation of $-7.4° \pm 11.9°$ and $-4.6° \pm 10.6°$ when compared to the Metelkin et al. (2007) and Kravchinsky et al. (2002)'s 155 Ma poles, respectively. However, a slight clockwise rotation of $13.5° \pm 8.5°$ was revealed when compared to the reference APWP pole at 160 Ma (Besse and Courtillot, 2002). Cogné et al. (2005) suggested that the clockwise rotation between the Trans-Baikal paleopoles and stable Europe was most probably caused by the shear movements along the Mongol–Okhotsk suture. This slight clockwise rotation between our 155 Ma poles and the reference APWP pole at 160 Ma might similarly was caused by shear movements along the Mongol–Okhotsk suture, which can involve a larger crustal area, covering the whole Trans-Baikal and northern Mongolia regions (Cogné et al., 2005), and may be covering the NCB today.

In terms of relative movements between the NCB and SIB since 155 Ma, this difference reflects a significant northward movement with respect to SIB, of the sampling site (40.8°N, 118.1°E) of $32.7° \pm 8.0°$, when compared to the Metelkin et al. (2007)'s 155 Ma pole. This result is similar to the Kravchinsky et al. (2002)'s pole ($34.8° \pm 7.2°$) coming from the same period and locality as that of Metelkin et al. (2007) and the 160 Ma pole ($34.6° \pm 6.2°$) of the reference APWP (Besse and Courtillot, 2002). Based on paleopoles coming from both sides of the Mongol–Okhotsk suture, Cogné et al. (2005) suggested that the Mongol–Okhotsk Ocean was still open at the end of the Jurassic and closed rapidly at 15 cm yr^{-1}. Our result

Table 4
The Middle and Late Jurassic paleomagnetic poles from NCB and adjacent regions.

Block	Age	Locality		*N*	VGP			Reference
		Long. (° E)	Lat. (° N)		Φ(° E)	λ(° N)	α95 (°)	
Eurasia	**160 Ma**			**15**	**144.0**	**72.5**	**5.0**	**Besse and Courtillot, 2002**
Eurasia	**150 Ma**			**10**	**159.9**	**75.0**	**6.6**	**Besse and Courtillot, 2002**
Siberia	**155 Ma**	**110.0**	**50.9**	**18**	**166.8**	**63.6**	**8.5**	**Metelkin et al., 2007**
Siberia	**155 Ma**	**112.0**	**51.8**	**9**	**161.0**	**64.4**	**7.0**	**Kravchinsky et al., 2002**
NCB	**J_2**	**109.3**	**36.2**	**6**	**232.8**	**74.3**	**5.0**	**Yang et al., 1992**
NCB	**J_2**	**116.0**	**31.6**	**17**	**254.7**	**72.9**	**6.4**	**Gilder and Courtillot, 1997**
NCB	J_3	119.2	35.8	5	31.1	81.0	9/13	Cheng and Fang, 1980
NCB	J_3	116.5	31.5	5	229.4	72.7	9/13	Cheng and Fang, 1980
NCB	J_3	117.5	35.8	10	215.6	74.2	4/5	Cheng and Fang, 1980
NCB	J_3	119.4	35.9	10	225.8	71.3	6/10	Lin, 1984
NCB	J_3	117.9	36.6	4	202.0	72.1	7/9	Lin, 1984
NCB	J_3	112.1	38.6	5	209.6	70.5	8.5	Fang et al., 1988
NCB	**J_3**	**116.0**	**31.6**	**10**	**222.8**	**74.4**	**5.9**	**Gilder and Courtillot, 1997**
NCB	**155 Ma**	**118.1**	**40.8**	**9**	**240.3**	**59.9**	**6.8**	**This study**
SCB	J_3	120.0	29.4	–	213.7	73.0	12.6	Lin et al., 1985
SCB	J_3	102.4	26.7	–	185.7	74.2	3.8	Huang and Opdyke, 1991
SCB	J_3	102.9	29.9	5	235.3	74.6	13.2	Enkin et al., 1992
SCB	J_3	106.7	31.8	11	236.0	64.7	7.0	Yokoyama et al., 1999
SCB	J_3	104.5	30.4	24	222.7	61.3	4.2	Yokoyama et al., 2001
SCB	**J_3**			**5**	**220.9**	**70.4**	**8.5**	**Mean result**
MOB	**J_3**	**117.5**	**49.5**	**8**	**254.8**	**73.0**	**7.8**	**Zhao et al., 1990**
MOB	**J_3**	**123.1**	**47.0**	**14**	**224.6**	**62.4**	**4.9**	**Zhao et al., 1990**
MOB	**J_3**	**117.4**	**51.7**	**6**	**275.9**	**73.3**	**6.3**	**Cogné et al., 2005**
MOB	**J_{2-3}**	**117.5**	**51.5**	**8**	**261.8**	**68.6**	**4.1**	**Kravchinsky et al., 2002**

Long., Lat. (Φ, λ) = longitude and latitude of sites (virtual geomagnetic pole); α95 = 95% confidence of the mean direction pole; N = number of sites used for pole determination; Bold data = poles used to discuss the paleoposition of relative block or areas.

suggests that the Mongol–Okhotsk Ocean was also open and might have been about 3000 km wide at 155 Ma. Also, deformation and continental crust thickening after the closure of the Mongol–Okhotsk Ocean contribute to this result. Geologically, crustal shortening prevailed along the China–Mongolia border areas during the Middle to early Late Jurassic in response to the continental collision of Siberia

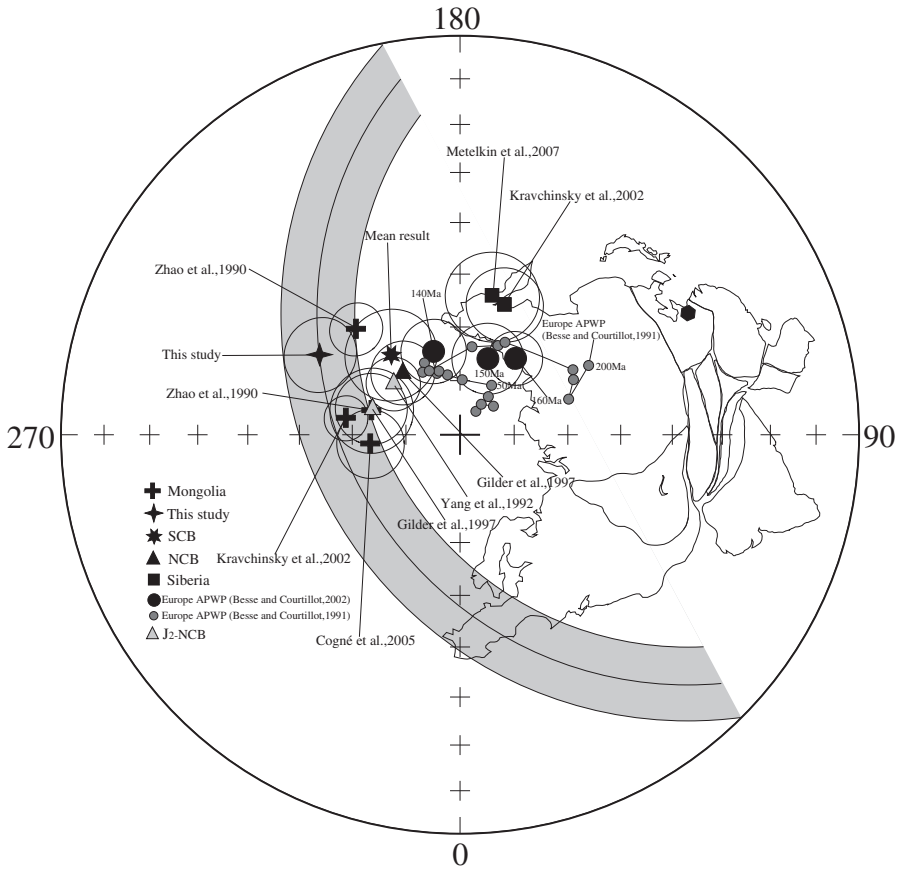


Fig. 6. Equal-area projections of palaeomagnetic poles in the Northern Hemisphere. Black hexagon: site locations; light gray area with bold black line: small-circle with its 95% confidence area passing through paleopoles, and centered on the site locations.

and the UEAB along the Mongol–Okhotsk suture (Davis et al., 2001; Yin and Nie, 1996; Zhao et al., 2004; Zorin, 1999).

Although this paper provides a reliable 155 Ma paleomagnetic pole for the North China Block obtained from the Tiaojishan Formation in Chengde area, the closure history of Mongol–Okhotsk Ocean requires more reliable paleomagnetic poles with accurate ages from the North China Block and other continents.

Acknowledgments

The authors express their deep gratitude for assistance in conducting the work and valuable advice from Andres Villavicencio, Mian Liu, Jean Pascal Cogné, and two anonymous reviewers. Paleomagnetic data were analyzed using R. Enkin's and J. P. Cogné's computer program package. We also thank Marie-Luce Chevalier for polishing the English. This study was funded by the Chinese National Natural Science Foundation (40602027, 41172177) and the Deep Exploration Technology and Experimentation Program of China (SinoProbe-08-01-03).

References

Badarch, G., Cunningham, W.D., Windley, B.F., 2002. A new terrane subdivision for Mongolia: implications for the Phanerozoic crustal growth of Central Asia. J. Asian Earth Sci. 21, 87–110.

Beijing Bureau of Geology and Mineral Resources, 1991. Regional geology of Beijing Municipality: Ministry of Geology and Mineral Resources. Geol. Mem. Ser. 1 (27) (598 pp).

Besse, J., Courtillot, V., 2002. Apparent and true polar wander and geometry of the geomagnetic field over the last 200 Myr. J. Geophys. Res. 107 (B11), 101029–101060.

Channell, J.E.T., Erba, E., Nakanishi, M., Tamaki, K., 1995. A Late Jurassic–Early Cretaceous timescale and oceanic magnetic anomaly block models. In: Berggren, W.A., Kent, D.V., Aubry, M.P., Hardenbol, J. (Eds.), Geochronology, Timescales, and Stratigraphic Correlation: SEPM (Society for Sedimentary Geology) Special Publication, 54, pp. 51–63.

Chen, A., 1996. A preliminary study on deformational mechanism of basement involved structure in the Yanshan Mountain: 30th International Geological Congress, Beijing (Abstracts2 291 pp).

Cheng, G., Fang, Z., 1980. The mode of motion of the Tancheng–Lujiang fault zone by preliminary paleomagnetic determinations. Seismolog. Geol. 2, 4.

Cogné, J.P., 2003. PaleoMac: a MacintoshTM application for treating paleomagnetic data and making plate reconstructions. Geochem. Geophys. Geosyst. 4 (1), 1007. doi:10.1029/2001 GC000227.

Cogné, J.P., Kravchinsky, V.A., Halim, N., Hankard, F., 2005. Late Jurassic–Early Cretaceous closure of the Mongol–Okhotsk Ocean demonstrated by new Mesozoic palaeomagnetic results from the Trans-Baïkal area (SE Siberia). Geophys. J. Int. 163, 813–832.

Cope, T.C., 2003. Sedimentary Evolution of the Yanshan Fold–Thrust Belt, Northeast China. Stanford University, Califonia, pp. 1–230.

Courtillot, V., Enkin, R., Yang, Z., 1994. Response to comment on "Paleomagnetic constraints on the geodynamic history of main Chinese blocks from the Permian to the present a review" by Enkin et al. (1992), by S.Nie and D.Rowley. J. Geophys. Res. 99, 18043–18048.

Cui, S.Q., 1999. On global Meso-Cenozoic intracontinental orogenesis and orogenic belts. Earth Sci. Front. 6 (4), 283–293.

Darby, B.J., Davis, G.A., Zheng, Y., 2001. Structural evolution of the southern Daqing Shan, Yinshan belt, Inner Mongolia, China. In: Hendrix, M.S., Davis, G.A. (Eds.), Paleozoic and Mesozoic Tectonic Evolution of Central and Eastern Asia: From Continental Assembly to Intracontinental Deformation: Geological Society of America Memoir, 194, pp. 199–214.

Davis, G.A., 2005. The late Jurassic "Tuchengzi/Houcheng" formation of the Yanshan Fold–Thrust Belt: an analysis. Earth Sci. Front. 12 (4), 328–344.

Davis, G.A., Zheng, Y.D., Wang, C., Darby, B.J., Zhang, C.H., Gehrels, G.F., 2001. Mesozoic tectonic evolution of the Yanshan fold and thrust belt, with emphasis on Hebei and Liaoning provinces, northern China. In: Hendrix, M.S., Davis, G.A. (Eds.), Paleozoic and Mesozoic Tectonic Evolution of Central and Eastern Asia: From Continental Assembly to Intracontinental Deformation: Geological Society of America Memoir, 194, pp. 171–197.

Enkin, R.J., Yang, Z.Y., Chen, Y., Courtillot, V., 1992. Paleomagnetic constrains on the geodynamic history of the major blocks of China from the Permian to the Present. J. Geophys. Res. 97, 13953–13989.

Fang, D., Guo, Y., Wang, Z., Tan, X., Fan, S., Yuan, Y., Tang, X., Wang, B., 1988. Tectonic implications of Triassic and Jurassic Paleomagnetic results from Ningwu basin, Shanxi Province. Kexue Tongbao 2, 133–135.

Fisher, R.A., 1953. Dispersion on a sphere. Proc. R. Soc. London, Ser. A 217, 295–305.

Gilder, S., Courtillot, V., 1997. Timing of the North–South China collision from new Middle to Late Mesozoic Paleomagnetic data from the North China Block. J. Geophys. Res. 102, 17713–17727.

Gradstein, F.M.G., Ogg, J.G., Smith, A.G., 2004. A new Geological Time Scale, with special reference to Precambrian and Neogene. Episodes 27 (2), 83–100.

Halim, N., Kravchinsky, V., Gilder, S., Cogné, J.P., Alexyutin, M., Sorokin, A., Courtillot, V., 1998. A Paleomagnetic study from the Mongol–Okhotsk region: rotated Early Cretaceous volcanics and remagnetized Mesozoic sediments. Earth Planet. Sci. Lett. 159, 133–146.

He, H., Pan, Y., Tauxe, L., Qin, H., Zhu, R., 2008. Toward age determination of the M0r (Barremian–Aptian boundary) of the Early Cretaceous. Phys. Earth Planet. Inter. doi:10.1016/j.pepi.2008.07.014.

Hebei Bureau of Geology and Mineral Resources, 1989. Regional Geology of Hebei Province, Beijing and Tianjin Municipality. Geological Publishing House, Beijing. (741 pp).

Huang, K., Opdyke, N.D., 1991. Paleomagnetism of Jurassic rocks from southwestern Sichuan and the timing of the closure of the Qingling suture. Tectonophysics N3, 40–49.

Kirschvink, J.L., 1980. The Least-Squares line and plane and analysis of paleomagnetic data. Geophys. J. R. Astron. Soc. 62, 699–718.

Kravchinsky, V.A., Cogné, J.P., Harbert, W., Kuzmin, M.I., 2002. Evolution of the Mongol–Okhotsk Ocean with paleomagnetic data from the suture zone. Geophys. J. Int. 148, 34–57.

Li, J.Y., 2006. Permian geodynamic setting of Northeast China and adjacent regions: closure of the Paleo-Asian Ocean and subduction of the Paleo-Pacific Plate. J. Asian Earth Sci. 26, 207–224.

Lin, J.L.,1984. The Apparent Polar Wander paths for the North and South China blocks, Ph.D. thesis, Univ. Calif. 248 pp.

Lin, J.L., Fuller, M., Zhang, W.Y., 1985. Preliminary Phanerozoic Polar Wander paths for the North and South China blocks. Nature 313 (7), 444–449.

Liu, J., Zhao, Y., Liu, X.M., 2006. Age of the Tiaojishan Formation volcanic in the Chengde Basin, northeast Hebei Province. Acta Petologica Sinica 22 (8), 2617–2630.

Lowrie, W., 1990. Identification of ferromagnetic minerals in a rock by coercivity and unblocking temperature properties. Geophys. Res. Lett. 17, 159–162.

Ma, X.H., McElhinny, M.W., Embleton, B.J.J., Zhang, Z.K., 1993. Permo-Triassic palaeomagnetism in the Emei Mountain Region, southwest China. Geophys. J. Int. 114, 293–303.

McElhinny, M.W., 1964. Statistics of the fold test in paleomagnetism. Geophys. J. R. Astron. Soc. 8, 338–340.

McFadden, P.L., McElhinny, M.W., 1990. Classification of the reversals test in palaeomagnetism. Geophys. J. Int. 103, 725–729.

Meng, Q.R., 2003. What drove late Mesozoic extension of the northern China–Mongolia tract? Tectonophysics 369, 155–174.

Metelkin, D.V., Gordienko, I.V., Klimuk, V.S., 2007. Paleomagnetism of Upper Jurassic basalts from Transbaikalia: new data on the time of closure of the Mongol–Okhotsk Ocean and Mesozoic intraplate tectonics of Central Asia. Russ. Geol. Geophys. 48, 825–834.

Nie, S., 1991. Paleoclimatic and paleomagnetic constraints on the Paleozoic reconstructions of South China, North China and Tarim. Tectonophysics 196, 279–308.

Nie, S., Rowley, D.B., 1994. Paleomagnetic constraints on the geodynamic history of the major blocks of China from the Permian to the present: comment. J. Geophys. Res. 99, 18035–18042.

Ogg, J.G., Agternerg, F.P., Gradstein, F.M., 2004. The Cretaceous Period. In: Gradstein, F.M., Ogg, J.G., Smith, A.G. (Eds.), A Geologic Time Scale 2004. Cambridge University Press, UK, pp. 344–383.

Pruner, P., 1992. Paleomagnetism and paleogeography of Mongolia in the Cretaceous, Permian and Carboniferous—final report. Phys. Earth Planet. Inter. 70, 169–177.

Watson, G.S., Enkin, R.J., 1993. The fold test in paleomagnetism as a parameter estimation problem. Geophys. Res. Lett. 20, 2135–2137.

Xiao, W.J., Windley, B.F., Hao, J., Zhai, M.G., 2004. Accretion leading to collision and the Permian Solonker suture, Inner Mongolia, China. Tectonics 22 (6), 1069. doi:10.1029/2002 TC001484.

Yang, Z.Y., Besse, J., 2001. New Mesozoic apparent polar wander path for south China: tectonic consequences. J. Geophys. Res. 106, 8493–8520.

Yang, Z., Courtillot, V., Besse, J., Ma, X.H., Xing, L., Xu, S., Zhang, J., 1992. Jurassic Paleomagnetic constraint on the collision of the North and South China Blocks. Geophys. Res. Lett. 19, 577–580.

Yin, A., Nie, S., 1996. A Phanerozoic palinspastic recostruction of China and its neighboring regions. In: Yin, A., Harrison, T.M. (Eds.), The Tectonic Evolution of Asia. Cambridge University Press, Cambridge (666 pp).

Yokoyama, M., Liu, Y.Y., Otofuji, Y., Yang, Z.Y., 1999. New Late Jurassic Paleomagnetic data from the northern Sichuan basin: implications for deformation of the Yangtze Craton. Geophys. J. Int. 139, 795–805.

Yokoyama, M., Liu, Y., Halim, N., Otofuji, Y., 2001. Paleomagnetic study of the Upper Jurassic rocks from the Sichuan basin: tectonic aspects for the collision between the Yangtze Block and the North China Block. Earth Planet. Sci. Lett. 193, 273–285.

Zhang, H.F., Sun, M., Zhou, X.H., Zhou, M.F., Fan, W.M., Zheng, J.P., 2003. Secular evolution of the lithosphere beneath the eastern North China Craton: evidence from Mesozoic basalts and high-Mg andesites. Geochim. Cosmochim. Acta 67, 4373–4387.

Zhang, S.H., Zhao, Y., Song, B., 2007. Carboniferous granitic plutons from the northern margin of the North China block: implications for a late Palaeozoic active continental margin. J. Geol. Soc. Lond. 164, 451–463.

Zhao, Y., 1990. The Mesozoic orogenies and tectonic evolution of the Yanshan area. Geol. Rev. 36, 1–13.

Zhao, X.X., Coe, R.S., 1987. Paleomagnetic constraints on the collision and rotation of north and south China. Nature 327, 141–144.

Zhao, X.X., Coe, R., Zhou, Y.X., Wu, H., Wang, J., 1990. New paleomagnetic results from northern China, collision and suturing with Siberia and Kazakhstan. Tectonophysics 181, 43–81.

Zhao, Y., Yang, Z.Y., Ma, X.H., 1994. An important transition in the development of east Asian tectonics. China. J. Geol. 29, 105–119.

Zhao, Y., Cui, S.Q., Guo, T., Xu, G., 2002. Evolution of a Jurassic basin of the Western Hills, Beijing, north China and its tectonic implications. Geol. Bull. China 21, 211–217.

Zhao, Y., Xu, G., Zhang, S.H., 2004. Yanshanian movement and conversion of tectonic regimes in East Asia. Earth Sci. Front. 11 (3), 319–328.
Zheng, Z., Kono, M., Tsunakawa, H., 1991. The Apparent Polar Wander path for the North China Block since the Jurassic. Geophys. J. Int. 104, 29–40.
Zonenshain, L.P., Kuzmin, M.I., 1997. Paleogeodynamics. Am. geophys. Un, Washington.
Zonenshain, L.P., Muzmin, M.I., Natapov, L.M., 1990. Mongol–Okhotsk fold belt, in Page, B.M., ed., Geology of the USSR: a plate tectonic synthesis. Am. Geophys. Union Geodyn. Ser. 21, 97–120.
Zorin, V.A., 1999. Geodynamics of the western part of the Mongolia–Okhotsk collisional belt, Trans-Baikal region (Russia) and Mongolia. Tectonophysics 306, 33–56.